Settlement & Imbalance Charge Guidance for ERCOT Exposure

Category: Reference | Model: VFIP-SETTLE-ERCOT-GUIDE

This document consolidates vendor firmware release notes, change ticket summaries, O&M incident report and postmortem, dispatch runbooks, and detailed settlement & imbalance-charge reconciliation guidance for ERCOT. It is intended for auditors, settlement analysts, operations, and RAG-enabled assistants.

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1. Overview & Purpose

This reference consolidates technical inputs required for accurate settlement and imbalance accounting for an ERCOT-exposed merchant fleet affected by the XG-440 firmware incident. It includes:

- Vendor firmware release notes and rollback guidance (v3.14.2 → v3.15.0).
- Change ticket rollouts and hotfixes with serial number ranges.
- O&M incident timeline and postmortem including curtailment windows.
- Dispatch runbooks mapping SCADA alarms to CMMS work orders.
- Detailed settlement and imbalance calculation guidance, mapping from analytical gold tables to settlement invoice line items, reconciliation steps, and an auditor checklist focused on reconciling a $1.42M exposure.

Intended use: internal audits, RAG knowledge base ingestion, settlement disputes, and corrective action planning.

2. Collection Context (XG-440 Firmware Incident)

Context: Firmware updates to inverters (XG-440 family) were deployed across multiple plant sites in late 2025. Some firmware versions introduced protective trip parameter changes and unexpected behavior that affected reported SCADA generation measurements vs. actual generation, producing large settlement deviations and imbalance penalties. This document ties the technical firmware changes to financial settlement effects.

Key impact window: 2025-11-20 00:00 UTC → 2025-12-08 23:59 UTC (includes initial rollout and hotfix timeframe).

Affected components

- Inverter model: XG-440, firmware family v3.14.x → v3.15.x
- SCADA telemetry: measured_mwh (per-interval net energy), status flags (INVERTER_TRIPPED), behavior of measured vs. calculated MWh
- Settlement data sources: gold_daily_generation, gold_revenue_variance (analytics tables); settlement_invoices_line_items (official invoice mapping)

3. Vendor Firmware Release Notes (v3.14.2 → v3.15.0)

This section provides the vendor's release notes excerpts and recommended actions. All items below are drawn from vendor-provided documentation in the incident pack.

v3.14.2 (baseline)

Baseline documented stable release prior to rollout.

Parameter/ Change	Description	Notes
VFD-Anti-Islanding (ais_enable)	Default: enabled. Controls anti-islanding detection thresholds.	Stable; no observed telemetry issues.
Trip Codes	TRIP-31: Overcurrent; TRIP-44: DC-domain overvoltage	Clear logging to SCADA; persisted status code in inverter700_status.
Telemetry batching	15s aggregation for SCADA measured_mwh computation.	Consistent with historical aggregation logic.

v3.15.0 (released)

Change highlights — introduced behavior that contributed to measurement discrepancies when certain protective thresholds were triggered.

Change	Impact	Recommended Rollback Condition
Parameter: trip_mode_aggressive = true	Increases sensitivity of protective trips (TRIP-31, TRIP-44). When enabled, inverter may issue a rapid trip and report "zero-output" for multiple telemetry intervals before properly clearing.	Rollback if unexpected net MWh discrepancy > 5% per plant within 24 hours or if simultaneous TRIP-31 counts spike > 10x baseline.
Change: measured_mwh_publish_delay = 30s	Delay introduced in publishing final measured MWh for an interval; in some cases SCADA collects intermediate zeroes.	Rollback if measured_mwh_time_series shows cluster of zeros at interval boundaries not matching inverter event logs.
Telemetry: status_code mapping changed	New status code "TRIP-31A" used for transient trips; SCADA parsers not updated may misclassify as permanent trip.	Rollback if >2% of "TRIP-31A" events are left uncleared in SCADA for > 1 hour.

> **Protective trip codes (observed):** TRIP-31, TRIP-31A, TRIP-44, TRIP-99 (firmware emergency stop). When TRIP-31/31A observed, inverter reports instantaneous output 0 kW until clear event logged.

Vendor rollback recommended to v3.14.2 if persistent reporting inconsistencies or large-scale curtailment not initiated by dispatch.

4. Change Ticket Summaries (2025-11-20 → 2025-12-08)

Summaries of vendor portal change tickets that correspond to the rollout and hotfix windows.

Ticket ID	Date	Scope (Plants / Serial Ranges)	Summary
CT-2025-11020-R1	2025-11-20	Plant A, Plant B (Serial XG440-000100 → XG440-000399)	Initial rollout v3.15.0. Enabled trip_mode_aggressive and adjusted publish delay. Planned rollout window 03:00–07:00 local.

CT-2025-11021-R2	2025-11-21	Plant C (Serial XG440-000400 → XG440-000599)	Extended rollout; included telemetry configuration change. Escalation flag set to monitor TRIP counts for 48h.
CT-2025-12008-HF1	2025-12-08	Plants A, B, C (hotfix targeted serials XG440-000100 → XG440-000599)	Hotfix v3.15.1 deployed to correct measured_mwh_publish_delay and status_code mapping (adds TRIP-31A handling). Advises field rollback to v3.14.2 if issues persist.

Rollout Notes

- Canary groups were scheduled but mixed manual and automated deployment methods caused partial adoption during the local night window.
- Monitoring thresholds to escalate were TRIP count increase > 5x baseline or total reported zero-output intervals > 1% of intervals across fleet.

5. O&M Incident Report & Postmortem

This O&M incident report includes timeline, curtailment windows, and validation steps performed by operations teams.

Executive summary

Between 2025-11-20 and 2025-12-08, multiple plants experienced anomalous zero-output telemetry and an increase in protective trips following vendor firmware v3.15.0 rollout. Net MWh recorded by SCADA during impacted intervals was lower than expected based on inverter output logs and internal generation models. Resulting imbalance charges and lost merchant revenue are estimated at approximately $1.42M across the impact window.

Day-by-day timeline (key events)

Date (UTC)	Event	Impact
2025-11-20 03:15	v3.15.0 deployed to Plant A canary group	TRIP-31 spike observed 03:20–04:00; SCADA reported zero-output for 12 intervals
2025-11-21 04:00	Rollout extended to Plant C	Similar pattern observed; operator alerted vendor
2025-11-22 → 2025-11-28	Ongoing elevated TRIP counts and mismatch between SCADA measured_mwh and inverter event logs	First preliminary estimate of imbalance exposure created
2025-12-01	Vendor confirms mapping change: TRIP-31A introduced	Proposes hotfix v3.15.1

2025-12-08 06:00	Hotfix deployed	Telemetry stabilized for majority of units by 2025-12-09 00:00

Curtailment windows and descriptions

For settlement purposes curtailment windows are defined as continuous periods of 5 or more settlement intervals (5-minute or 15-minute depending on meter) where reported measured_mwh is 0 kW and inverter status indicates no commanded dispatch curtailment.

- Example window: 2025-11-20 03:20–04:05 (15-minute intervals: 03:15–03:30, 03:30–03:45, 03:45–04:00)
- Curtailed intervals flagged where SCADA status_flag != "DISPATCH_CURTAIL" and inverter_event_log shows TRIP-31/31A/44

Validation steps performed

1. Cross-check inverter event logs (local device logs) vs. SCADA measured_mwh time series.
2. Compare gross revenue expected (forecast * LMP) vs. gross_revenue_usd computed from gold_revenue_variance.
3. Flag intervals where scheduled_mwh > 0 and measured_mwh == 0 without a dispatch curtail instruction.
4. Engage vendor for root cause — parameter trip_mode_aggressive and measurement publish delay identified.

Key findings

- Telemetry lag and transient trip classification caused SCADA to record zero-output for full settlement intervals when real generator output resumed quickly.
- These misreported intervals produced imbalance charges because deviations from scheduled dispatch were reported to ERCOT as negative generation.

6. Dispatch Runbooks: SCADA Alarms → CMMS Work Orders

Standardized runbooks to convert SCADA alarms into corrective work orders. Use this to ensure corrective actions and evidence capture for settlement dispute support.

Runbook: TRIP-31 / TRIP-31A spike (protective trip)

1. Alert: SCADA triggers ALARM: INVERTER_TRIP spike > 5 events / 15 minutes.
2. Investigate: Pull inverter local event logs for matching timestamps. Query the inverter for firmware_version and status_code list.

3. Immediate action: If firmware_version in [v3.15.0], mark as *firmware-suspect* and submit CMMS corrective order template "CMMS-WO-FW-TRIP-INV".
4. CMMS WO fields (template): plant_id, inverter_serial, event_count, first_event_ts, last_event_ts, firmware_version, actions_taken (rollback_requested/patch_applied), attachment: inverter_event_log.csv
5. Escalation path: Open vendor support case; include signed event logs and SCADA snapshot for affected intervals.

CMMS Work Order Templates (examples)

```
WO_TEMPLATE: CMMS-WO-FW-TRIP-INV
Fields:
  - plant_id
  - inverter_serial_start
  - inverter_serial_end
  - firmware_version
  - event_codes (e.g., TRIP-31, TRIP-31A, TRIP-44)
  - event_count
  - start_interval (UTC)
  - end_interval (UTC)
  - assigned_to
  - priority (P1/P2/P3)
  - corrective_action (rollback/firmware-patch/replace)
  - attachments (inverter_logs.csv, scada_snapshot.json)
```

Evidence capture for settlements

For each flagged interval include:

- SCADA measured_mwh time series (native resolution).
- Inverter event log entries with exact timestamps and status codes.
- Dispatch setpoint history (scheduled_mwh) and any manual curtail instructions.
- Change ticket ID and firmware version applied.

7. Settlement & Imbalance Charge Guidance (ERCOT)

This section explains how merchant revenue, imbalance charges, and penalties are typically calculated and where mismatches occur due to telemetry issues.

Key settlement concepts

- Scheduled MWh: energy scheduled with ERCOT (per settlement_interval_id)
- Measured MWh: energy measured and reported (SCADA/settlement meter) for the interval
- Deviation (ΔMWh): measured_mwh - scheduled_mwh
- Real-Time LMP (RT_LMP_usd_per_mwh): the price used to value

deviations in RT imbalance charges (simplified guidance)
- Gross revenue: revenue from scheduled and dispatched energy at applicable LMPs
- Imbalance charge: typically ΔMWh * RT_LMP (direction and market rules impact settlement netting and potential specific penalty rates)

Standard equations (simplified)

Notation:

```
Let:
  scheduled_mwh = S (MWh)
  measured_mwh  = M (MWh)
  deviation_mwh = D = M - S
  RT_LMP_usd_per_mwh = P ($/MWh)
  settlement_interval_id = unique interval ID (e.g., 2025-11-20T03:15Z)
  gross_revenue_usd = revenue associated with scheduled delivery (if
applicable)
```

Imbalance Charge (simplified conventional calculation):

```
imbalance_charge_usd = D * P
If D is negative (measured < scheduled), imbalance_charge will be
negative revenue to the generator (i.e., a charge).
```

Note: ERCOT uses more granular market settlement constructs (Net Settlement, Secondary mismatch adjustments, uplift allocations). The simplified formula above is used for variance tagging; final invoice line items depend on settlement invoice schema.

Common causes of imbalance exposure in this incident

1. Telemetry zero-reporting for full settlement intervals while physical production resumed mid-interval — measured_mwh undercount.
2. Status code misclassification causing intervals to be treated as forced outage rather than transient trip.
3. Delayed publish of final meter aggregation producing intermediate zeroes into settlement feed.

8. Data Mapping: gold_* Tables → settlement_invoices_line_items

This section documents the mapping from analytics "gold" tables to the official settlement invoice line item fields. Use this as the canonical mapping for reconciliations and RAG ingestion.

Gold table schema excerpts (relevant fields)

gold_daily_generation	Type	Description

settlement_interval_id	string / timestamp	Unique settlement interval identifier (UTC)
plant_id	string	Internal plant identifier
measured_mwh	decimal	Measured generation for interval (MWh)
meter_id	string	Settlement meter identifier

gold_revenue_variance	Type	Description
settlement_interval_id	string / timestamp	Same interval ID
scheduled_mwh	decimal	Committed or cleared schedule (MWh)
rt_lmp_usd_per_mwh	decimal	Real-time LMP for interval
gross_revenue_usd	decimal	Revenue calculated for the interval (pre-adjustments)
variance_reason_code	string	Classification for variance (e.g., FIRMWARE-TRIP, SCADA-LAG, CURTAILMENT)

Settlement invoice model: settlement_invoices_line_items (canonical fields)

Field	Description	Source mapping
invoice_id	Settlement invoice identifier	Derived from settlement batch
line_item_id	Line item identifier	Computed per interval
settlement_interval_id	Interval timestamp	gold_daily_generation.settlement_interval_id
measured_mwh	Measured generation	gold_daily_generation.measured_mwh
scheduled_mwh	Scheduled generation	gold_revenue_variance.scheduled_mwh
deviation_mwh	Measured - Scheduled	computed D = M - S
rt_lmp_usd_per_mwh	Real-time price	gold_revenue_variance.rt_lmp_usd_per_mwh
imbalance_charge_usd	Charge or credit	deviation_mwh * rt_lmp_usd_per_mwh
gross_revenue_usd	Gross revenue assigned	gold_revenue_variance.gross_revenue_usd
variance_reason_code	Classification	gold_revenue_variance.variance_reason_code

ETL mapping pseudocode

```
-- Simplified SQL-style transformation (pseudocode)
SELECT
  s.invoice_id,
  concat(s.invoice_id,'-',g.settlement_interval_id) AS line_item_id,
  g.settlement_interval_id,
  g.plant_id,
  g.measured_mwh,
  r.scheduled_mwh,
  (g.measured_mwh - r.scheduled_mwh) AS deviation_mwh,
  r.rt_lmp_usd_per_mwh,
  (g.measured_mwh - r.scheduled_mwh) * r.rt_lmp_usd_per_mwh AS
imbalance_charge_usd,
  r.gross_revenue_usd,
  r.variance_reason_code
FROM gold_daily_generation g
JOIN gold_revenue_variance r
  ON g.settlement_interval_id = r.settlement_interval_id
LEFT JOIN settlement_invoices s
  ON s.settlement_date = DATE(g.settlement_interval_id)
WHERE g.plant_id IN ('Plant A', 'Plant B', 'Plant C');
```

9. Reconciliation Procedure & Examples

This section describes step-by-step reconciliation procedures to validate net MWh and compute imbalance impact. It contains worked examples demonstrating how deviations create charges.

Prerequisites

- Access to gold_daily_generation, gold_revenue_variance, settlement_invoices_line_items, inverter event logs, and SCADA snapshots.
- Definitions table with settlement_interval granularity (15-minute vs. 5-minute).
- Historical baseline TRIP counts and baseline measured_mwh distribution for each plant.

Step-by-step reconciliation to validate net MWh

1. Extract affected settlement_interval_id list for impact window: 2025-11-20 → 2025-12-08.
2. For each interval, pull:
    - gold_daily_generation.measured_mwh (M)
    - gold_revenue_variance.scheduled_mwh (S)
    - gold_revenue_variance.rt_lmp_usd_per_mwh (P)
    - inverter_event_log entries overlapping interval

- SCADA status flags (DISPATCH_CURTAIL, MANUAL_TRIP, etc.)

3. Compute deviation D = M - S and per-interval imbalance_charge_usd = D * P.
4. Sum per-interval imbalance charges to produce net imbalance exposure for the period:

```
net_imbalance_usd = SUM_over_intervals( (M_i - S_i) * P_i )
net_deviation_mwh = SUM_over_intervals( M_i - S_i )
```

5. Cross-check measured_mwh vs. inverter instantaneous power logs (kW samples). If inverter logs show non-zero power during interval, create a variance_reason_code "SCADA-LAG" or "FIRMWARE-TRIP".
6. Generate variance evidence package: for each interval with variance_reason_code in (FIRMWARE-TRIP, SCADA-LAG), attach inverter logs, SCADA snapshot, and change ticket ID.

Worked example (single interval)

Interval: 2025-11-20T03:15Z (15-minute interval)

Field	Value
scheduled_mwh (S)	1.200 MWh (average 4.8 MW over 15 minutes)
measured_mwh (M)	0.000 MWh (SCADA recorded zero for interval)
rt_lmp_usd_per_mwh (P)	$45.00 / MWh
deviation (D = M - S)	-1.200 MWh
imbalance_charge_usd	-1.200 * $45.00 = -$54.00 (charge)
evidence	Inverter log shows transient trip 03:17–03:17:30 and full output restored 03:17:45; SCADA published intermediate zeros due to publish delay.

Aggregate Example (short window)

3 affected intervals with the following (S, M, P):

Interval	S (MWh)	M (MWh)	P ($/MWh)	D (MWh)	imbalance ($)
03:15	1.200	0.000	45.00	-1.200	-54.00
03:30	1.150	0.300	47.00	-0.850	-39.95
03:45	1.100	1.050	46.00	-0.050	-2.30
Total	**3.450**	**1.350**		**-2.100**	**-96.25**

Analyst action: group intervals by variance_reason_code and attach evidence. If vendor acknowledges firmware-caused measurement undercounting, submit settlement variance claim package to ERCOT per protocol within dispute window.

Troubleshooting common anomalies

- Symptom: clusters of zero reported M across intervals despite inverter logs showing non-zero instantaneous power — Root cause: measured_mwh_publish_delay or status_code misclassification. Resolution: vendor hotfix or rollback; compute corrected M by integrating inverter instantaneous power samples and produce variance claim.
- Symptom: measured_mwh occasionally higher than scheduled due to SCADA duplication — Root cause: telemetry duplication or meter aggregation errors. Resolution: deduplicate entries; validate against meter IDs and physical meter readings.

10. Auditor Checklist — Reconcile $1.42M Lost Revenue & Penalties

Use this checklist to validate the $1.42M exposure claim, preparing evidence for settlement dispute and insurance or vendor claim review.

1. Define impact window and impacted plant list:
    - Impact window: 2025-11-20 00:00 → 2025-12-08 23:59
    - Plants: Plant A, Plant B, Plant C (cross-check serial ranges XG440-000100 → XG440-000599)
2. Data extraction:
    - Extract gold_daily_generation and gold_revenue_variance for the window.
    - Extract settlement_invoices_line_items and invoice PDFs for relevant settlement batches.
3. Compute baseline expected revenue:
    - For each interval compute expected_revenue = scheduled_mwh * RT_LMP_usd_per_mwh
    - Aggregate to period gross_expected_revenue
4. Compute observed revenue and imbalance:
    - observed_revenue = measured_mwh * settlement_price (as per invoice)
    - imbalance_usd = SUM((measured_mwh - scheduled_mwh) *

RT_LMP)

5. Attribute variance items:
    - For each negative deviation, attach inverter log showing that inverter produced energy within interval (proof of measurement undercount)
    - Tag variance_reason_code (FIRMWARE-TRIP / SCADA-LAG / CURTAILMENT)
6. Vendor evidence:
    - Include change ticket IDs (CT-2025-11020-R1, CT-2025-11021-R2, CT-2025-12008-HF1)
    - Include vendor release notes snippet indicating trip_mode_aggressive and measured_mwh_publish_delay
7. Calculate corrective revenue (remediation estimate):
    - Corrected_M_i = integrate(inverter_kW_samples over interval) / 1000 (MWh)
    - corrected_imbalance_usd_i = (Corrected_M_i - S_i) * P_i
    - sum corrected_imbalance_usd over impacted intervals = remediation_value
8. Reconcile to $1.42M:
    - Sum invoice imbalance charges and compare to remediation_value; document the delta.
    - Check for ancillary charges or settlement adjustments affecting final invoice (uplift, netting, penalties).
9. Prepare dispute package:
    - For ERCOT: include interval list, corrected M calculations, inverter logs, vendor change tickets, and operator notes.
    - Files: evidence_manifest.csv listing supporting files keyed by settlement_interval_id
10. Final sign-off:
    - Operations Manager approves evidence package.
    - Legal and Finance approve submission.

Sample calculation to reach $1.42M

Illustrative aggregation (rounded):

Category	Amount (USD)
Direct imbalance charges (sum of interval charges)	$1,050,000.00

Lost gross revenue (unsettled scheduled deliveries)	$300,000.00
Late-payment fees / interest / ancillary penalties	$70,000.00
Total exposure	**$1,420,000.00**

> Action: Recompute each line using corrected measured M values. If corrected values reduce direct imbalance charges by >$1.42M, escalate to vendor for indemnity or file dispute with ERCOT per settlement rules.

11. Appendices: Codes, Field Glossary, Change Ticket Log

Protective Trip Codes (observed)

Code	Meaning	Operational Response
TRIP-31	Overcurrent protection trip	Inspect inverter, review protection curves, collect logs
TRIP-31A	Transient Overcurrent (introduced v3.15.0)	Check for rapid clear; if persistent, classify as firmware issue
TRIP-44	DC overvoltage	Investigate DC side, string-level issues
TRIP-99	Emergency stop (firmware)	High priority; immediate field action

Field Glossary (selected)

Field	Description
settlement_interval_id	Canonical interval identifier aligned with ERCOT settlement interval (UTC)
measured_mwh	Measured energy within interval in MWh (from gold_daily_generation)
scheduled_mwh	Cleared or committed schedule in MWh (gold_revenue_variance)
gross_revenue_usd	Calculated base revenue for interval before adjustments
variance_reason_code	Classification tag for anomalies; used in invoice line_item comments

Change ticket log (complete)

```
CT-2025-11020-R1  : 2025-11-20 : v3.15.0 rollout (Plant A/B) : Serial
XG440-000100 → 000399
CT-2025-11021-R2  : 2025-11-21 : v3.15.0 rollout (Plant C)   : Serial
XG440-000400 → 000599
```

```
CT-2025-12008-HF1 : 2025-12-08 : Hotfix v3.15.1 deployed     : Target
all serials above
```

12. Contacts & Next Steps

If you are preparing a settlement variance claim or conducting an audit, coordinate with the following roles:

Role	Responsibility	Contact (internal)
Operations Lead	Gather SCADA snapshots & inverter logs	ops.lead@example.com
Settlement Analyst	Compute interval charges & prepare invoice evidence	settlement.team@example.com
Vendor Liaison	Coordinate firmware actions & evidence from vendor	vendor.liaison@example.com
Legal / Compliance	Review dispute submissions	legal@example.com

Recommended immediate next steps

1. Complete the auditor checklist and produce evidence packages for top 50 impacted intervals by dollar amount.
2. Request vendor confirmation of firmware cause (written) and remediation plan.
3. Prepare and file settlement dispute to ERCOT with complete evidence within dispute window.
4. Plan firmware rollback schedule for units still on v3.15.0 not corrected by hotfix.

Document version: VFIP-SETTLE-ERCOT-GUIDE v1.0 — Generated for RAG ingestion and internal audit workflows. Maintain copies of all raw evidence files (inverter logs, SCADA exports, invoice PDFs) with robust access controls.

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This document is intended for internal use. The settlement equations are simplified illustrations; use final ERCOT invoice logic for binding reconciliation.